TOMMY DO

Data Center & Network Operations Engineer

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SUMMARY

Data Center & Network Operations Engineer with 2+ years of hands-on experience designing, operating, and automating production multi-tenant data center networks. Proven track record building Python and TypeScript internal automation platforms, managing high-throughput BGP/OSPF topologies, optical DWDM transport, and multi-vendor spine-leaf architectures. Leverages AI-assisted tooling (LLMs, Copilot, RAG) to accelerate network scripting, Jinja2 template generation, documentation, and operational workflows.

TECHNICAL SKILLS

Networking & Protocols: BGP, OSPF, IPv4/IPv6, Spine-Leaf Architecture, VLAN/VRF, RPKI, DNS, TCP/UDP, 40G/100G Optics, DWDM, Dark Fiber Ring, L1–L7 Troubleshooting

Automation & Scripting: Python, TypeScript, Bash, Ansible, Jinja2, FastAPI, REST APIs, Git, SvelteKit, AI/LLM Workflows (Ollama, RAG, Copilot)

Security & Compliance: FortiGate, FortiEMS, FortiNAC (802.1X), Network ACLs, Firewall Rules, SOC 2 Evidence Collection

Observability & Systems: Prometheus, Grafana, SNMP, gNMI, Syslog, Wireshark/Packet Analysis, Linux (Ubuntu, Debian, AlmaLinux), MySQL, IPMI

PROFESSIONAL EXPERIENCE

NOC / Network Engineer 2024 – Present

Open Colo — Data Center Operations — San Jose, CA

- **Allocation portal —** architected and shipped an internal full-stack allocation portal using TypeScript/SvelteKit, Python/FastAPI, and Ansible, replacing legacy PHP systems; reduced operational ticket processing time by 40%.
- **Config backups —** built automated daily network switch configuration backups with change-hashing, Git versioning, and diff tools across 100+ network devices, eliminating manual change tracking and improving audit accuracy.
- **Customer network design —** designed and deployed redundant customer network architectures, including spine-leaf topologies supporting high-density 40-server pod deployments with dedicated IPMI and redundant uplinks.
- **Automation tooling —** developed Python and Bash automation tools for automated RPKI configuration, geofeed allocation, and LOA PDF generation, cutting turnaround time on customer engineering requests from hours to minutes.
- **Observability —** integrated Prometheus and Grafana observability stacks leveraging SNMP and gNMI telemetry to monitor multi-ISP peer connectivity, with automated alerting for link degradation and packet drops.
- **Optical & physical layer —** administer DWDM transport systems and optical dark fiber rings connecting multi-site data centers; manage physical layer infrastructure including fiber splicing, OTDR/power testing, and cross-connects.
- **Security & compliance —** manage FortiGate firewall policies and FortiNAC 802.1X endpoint compliance; generate quarterly technical audit evidence for SOC 2 compliance.
- **AI-assisted automation —** leverage local LLMs (Ollama/Qwen) and AI-assisted tooling to accelerate Python scripting, auto-generate Jinja2 network templates, and synthesize post-incident runbooks.

Election Support Lead & Technical Specialist Nov 2020 – Sep 2021

Santa Clara County Registrar of Voters — Santa Clara, CA

- Managed a team of operational staff, ensuring strict compliance with procedural and legal requirements.
- Triaged and resolved hardware, software, and connectivity issues on voting devices under tight SLAs, ensuring 100% operational uptime across precinct equipment.

PERSONAL PROJECTS

Homelab & Infrastructure Environment

- Designed and deployed a self-hosted Linux container environment running Jellyfin, Paperless-ngx, and Pi-hole DNS filtering over custom Tailscale VPN overlay networks.

Paperfolio (Trading Simulator)

- Built a real-time portfolio management application using SvelteKit, FastAPI, SQLite, and external REST APIs to model trading logic and execution signals.

EDUCATION

California State University, East Bay — B.S. Computer Science 2022

Hayward, CA — Relevant Coursework: Computer Networks, Operating Systems, Database Architecture, Data Structures & Algorithms